Efficient CIFAR-100 Modelling

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Abstract

This paper overviews two CIFAR-100 models, a compact classifier and generative model. The classifier was inspired by the MobileNet series [1], which was designed to work with limited mobile capacity. It incrementally builds on a slimmed-down version, implementing select parts of MobNetv2 [2] and MobNetv3 [3] to strengthen accuracy. The generator is a projection-based cGAN, with inspiration and subsequent updates based on the DCGAN architecture [4] and subsequent iterative improvements. Together, they explore cutting-edge deep learning techniques under tight constraints.

Part 1: Classification

1 Methodology

Given the 100k parameters and 10k optimisation budget, I based the classifier on the MobileNet family. MobileNets were designed for resource-constrained environments and achieve efficiency by factorising a standard convolution into a depthwise spatial filter followed by a 1×1 pointwise projection, dramatically reducing parameters whilst maintaining representational capacity. Formally, a standard 3×3 convolution with C_{in} input channels and C_{out} output channels requires $9C_{\text{in}}C_{\text{out}}$ parameters, whereas its depthwise–separable counterpart uses only $9C_{\text{in}} + C_{\text{in}}C_{\text{out}}$. This reduction supports strong accuracy–parameter trade-offs, making MobileNet architecture ideal for this coursework.

To being, I implemented a tiny version of MobileNetV1 [1] using a 3x3 stem convolution with stride=1 to extract low-level features, followed by a sequence of depthwise-separable convolutions forming the core feature-extraction process. In line with MobileNetV1, all depthwise filters operated on channels independently, before the pointwise layer mixes information cross-channels. I applied global average pooling before the classifier head and introduced a width multiplier α to scale base channels down so the model fit within the 100k parameter limit. This already offered a solid baseline, comfortably clearing 40% test accuracy.

I then expanded with an inverted-residual structure from MobileNetV2 [2], reorganising the depthwise-separable principle around a narrow→wide→narrow sequence. I expanded the channels with 1x1 pointwise convolutions, applied depthwise convolutions and projected back to a linear bottleneck via a second pointwise layer. As discovered in MobileNetV2, these linear bottlenecks preserve information by avoiding non-linear activation collapses in the compressed space. Residual connections were used where input and output shapes matched. An expansion and reduction factor were added as configurable per-block parameters to control intermediate width and ReLU6 replaced ReLU as the non-linearity inside bottleneck layers as it provided greater stability under the high channel compression. This bought test accuracy above 55%.

Finally I incorporated a subset of MobileNetV3-inspired elements [3], recognising the full architecture wasn't suitable under constraints. I utilised a frugal ECA (Efficient Channel Attention) class based on MobileNetV3's use of squeeze-and-excite that applied global average pooling followed by a one-dimensional convolution over channels to model local cross-channel dependencies at lower cost. This was applied only to mid-to-late level inverted residual blocks to preserve early stage efficiency and avoid disrupting low-level feature extraction. In parallel, I generalised inverted-residual blocks to support per-block expansion and reduction factors alongside configurable depthwise convolutions in deeper layers, allowing me to optimise parameter spending whilst enhancing the receptive field. The activation

function was made a hyperparameter so HardSwish could be experimentally used where its smoother peicewise-linear behaviour improved optimisation and ReLU6 retained within linear bottleneck layers due to its bounded output range and stability on small channel widths.

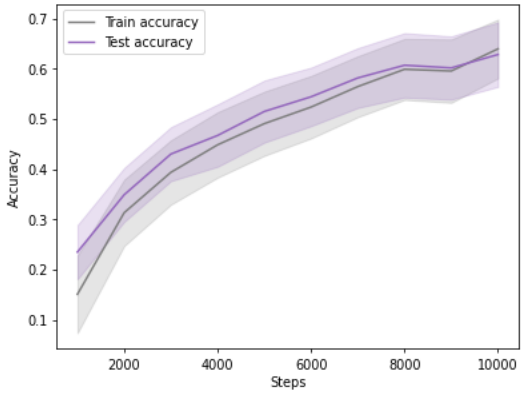
Cross-entropy loss was used throughout, and the original Adam optimiser was replaced with stochastic gradient descent (SGD) with momentum which gave smoother convergence and a ˜3% improvement in test accuracy. SGD also interacted more effectively with cosine annealing. Training followed a two-phase schedule: a cosine decay over the first 8k steps followed by a reduced learning rate for the last 2k steps to stabilise convergence and tighten the train-test gap. The resulting model is now a bespoke sub-100k-parameter MobileNetV2/V3 variant using depthwise-separable blocks throughout, calculated use of ECA channel attention in deeper layers, application of a global average pooling and a singular, fully-connected and dropout-regularised classifier head. Combined, this yielded 64% test accuracy.

2 Results

The network has 99344 parameters.

It attains 64.0% training accuracy and also 62.9% testing accuracy at 10,000 optimisation steps, demonstrating a significant increase on the provided baseline model. The train-test gap is ˜1%, indicating stable generalisation without overfitting. Variance is low, indicating high stability. (train loss: 1.255, train acc: 0.640±0.059, test acc: 0.629±0.064).

Here is the training graph:



3 Limitations

Limitations include the 100k parameter cap, which necessitated a significant accuracy trade-off: in line with MobileNetV1's findings, I found increasing the width multiplier α beyond 0.8 gave significant accuracy gains but greatly overshot parameter budget, pointing to capacity as a blocker. This trade-off was anticipated, so α and base-channel count were treated as primary hyperparameters, pushed as high as possible whilst keeping the model marginally below 100k parameters, yet this did impose a hard performance cap. Secondly, although mixup [5] and cutmix [6] are common in CIFAR-100-related literature, they consistently reduced accuracy by around 5%. This again points to limited capacity as the culprit, with the network likely lacking capacity to exploit more challenging augmented distributions.

Thirdly, the 10k optimisation step budget restricted how far the methods can be tuned, and thus called for careful two-step tuning of the learning rate to minimise loss. Finally, the MobileNetV3 architecture is incompatible with constraints: even "small" MobileNetV3 variants exceed $>$ 2 million parameters and assume 224x224 inputs with repeat stride=2 downsampling, which would collapse the CIFAR100 32x32 images too aggressively. Thus, my hand-crafted model only selectively uses 5x5 depthwise kernels in deeper layers and drastically cuts the expansion factor to keep within constraints, further lowering the accuracy ceiling.

Part 2: Generative model

4 Methodology

Under the 50k optimisation steps and 1M parameter budget, the primary challenge was utilising capacity on stability first, then architectural tweaks. I therefore adopted a similar methodological strategy Part 1, establishing a simple but stable baseline and then applying incremental, research-backed upgrades to address specific optimisation failures. Each modification was introduced in isolation and evaluted with ablations, kept only when it actually demonstrated clearer image quality.

My initial baseline was a minimal DCGAN-style generative model [4], which I chose for its well-established convolutional inductive bias for generating images. Mathematically, it consists of a generator $G(z, y)$ which maps a latent vector $z \sim \mathcal{N}(0, I)$, and class label y to an image, alongside a discriminator $D(x, y)$ which assigns a real-valued score to an image–label pair. Training uses a hinge-based adversarial objective [7], and the losses are:

$$\mathcal{L}_D = \mathbb{E}_{x,y}\big[\max(0, 1 - D(x, y))\big] + \mathbb{E}_{z,y}\big[\max(0, 1 + D(G(z, y), y))\big], \tag{1}$$

$$\mathcal{L}_G = -\mathbb{E}_{z,y}\big[D(G(z, y), y)\big]. \tag{2}$$

This moved the discriminator from a probabilistic classifier to a margin-based critic, which advantageously preserves gradients when the discriminator becomes overly confident. To further stabilise training I applied spectral normalisation to the discriminator and generator [7], which rescales parameters for each weight matrix W as:

$$\bar{W} = \frac{W}{\sigma(W)}, \tag{3}$$

where $\sigma(W)$ denotes the largest singular value of the linear transformation induced by W. This constrains an upper bound on the discriminator's Lipschitz constant to prevent uncontrolled sensitivity alongside maintaining informative gradients.

The next step was introducing residual blocks in both generator and discriminator to improve optimisation with shortcut connections. Here, residual blocks use the mapping:

$$h_{l+1} = h_l + F(h_l), \tag{4}$$

to ensure gradients propagate straight through the identity connection.

Alone these residual connections actually initially degraded sharpness, so I removed batch normalisation in generator upsampling blocks (but retained later in the generator before output convolution). Formally, batch normalisation is defined as:

$$\hat{h} = \frac{h - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \qquad \mathrm{BN}(h) = \gamma\hat{h} + \beta, \tag{5}$$

[8] and at small batches this can introduce noise and generate less sharp images [9].

Next, class conditioning was applied to both generator and discriminator to capture CIFAR-100's multi-modal structure [10]. In the discriminator, conditioning was implemented with a projection-based formulation [11] defined as:

$$D(x, y) = f(x) + \langle h(x), e(y)\rangle, \tag{6}$$

where $h(x)$ is the pooled feature representation and $e(y)$ is a learned class embedding. This injects class information straight into the decision function so semantic attributes actively shape the discriminator boundary.

The final gains were through refining training dynamics. Firstly I increased the discriminator-to-generator update ratio using a TTUR-inspired (two time-scale update rule) asymmetric update schedule [12], which pushed the discriminator to be sufficiently optimised to provide informative gradients (although it did reduce total training steps).

An EMA (exponential moving average) of generator parameters was used for evaluation and sampling [13], defined by:

$$\theta_{\text{EMA}}^{(t)} = \alpha\theta_{\text{EMA}}^{(t-1)} + (1-\alpha)\theta^{(t)}. \tag{7}$$

This smooths short-term adversarial oscillations without affecting training dynamics. I also introduced a warmup period to reduce the intense effect of smoothing during the chaotic early phase.

Finally, I trialled (and removed) mild truncation on the latent space to limit extreme latent deviations (consistent with [14]), which increased diversity but reduced realism and FID.

5 Results

The network has 985,796 parameters and achieves an FID of 38.02 against the CIFAR-100 test dataset. It was trained for 50,000 optimisation steps.

The results show generation of novel, semi-realistic images. A random batch of non-cherry picked samples looks like this (left) and linear interpolants between points in the latent space look like this (right):



The mean LPIPS distance between a 64-sample batch and its nearest neighbor in the 10000 training images (using a fixed random seed) was 0.0985, indicating limited memorisation and genuine sample novelty under the constraints [15].

6 Limitations

Despite big gains from the baseline, the final model was still quite sensitive to both random initialisation and hyperparameter choices, which does show the inherently unstable nature of adversarial optimisation under such strict step constraints. Although stabilisation techniques like hinge loss and spectral normalisation, TTUR and EMA did reduce variance, they still didn't completely remove it.

The biggest limits were the constrained image resolution and optimisation budget, which placed a hard ceiling on how many fine-grained details could be learned. There are a few enhancements which are just ineffective at this small of a resolution (e.g. style mixing and path length regularisation were explored but non-functional [7]), suggesting major new gains are most likely to come from a more generous optimisation budget.

Another major limitation is my own ability to judge output quality, inherently adding human bias when deciding which architectural tweaks add realism. To improve, a more exhaustive analysis of sample uniqueness could be added (explored in [14]), but this fell outside scope.

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